COMP5211 Introduction to Database Systems - Course Syllabus

Course description. This course introduces the design, implementation and use of relational database systems for students entering the MSc programme with no prior database background. It begins with the relational model as a mathematical object, moves through schema design and normalisation, and then covers the SQL language in enough depth that students can express realistic analytical queries without help. The second half of the term turns to what happens underneath the query: storage layout, indexing structures, transaction management and the guarantees a database makes when several users write at the same time. The course is deliberately practical. Every concept introduced in lecture is exercised the same week against a live PostgreSQL instance provided to each student, because experience shows that students who only read about isolation levels do not retain them.

Learning outcomes. On successful completion of this course a student will be able to: design a normalised relational schema from an informal description of a business domain and justify the normal form reached; write correct SQL queries involving joins, aggregation, subqueries and window functions; read an execution plan and explain why the planner chose a particular access path; reason about the correctness of concurrent transactions using the ACID properties and the standard isolation levels; and evaluate the trade-offs between relational and non-relational storage for a stated workload. These outcomes are assessed cumulatively rather than in isolation, so the final examination assumes everything earlier in the term.

Prerequisite. Students are expected to be comfortable with elementary discrete mathematics, in particular sets, relations and functions, and to be able to write and debug a small program in any imperative language. No prior exposure to SQL or to any particular database product is assumed or required. Students who have not programmed for several years should contact the instructor during the first week.

Assessment, Policies and Resources

Grading. The final grade is composed of four written assignments worth forty per cent in total, a midterm examination worth twenty per cent, a term project worth fifteen per cent, and a final examination worth twenty-five per cent. Assignments are released on Monday and are due at the end of the following week. Late work loses ten per cent of its available marks per calendar day and is not accepted more than five days after the deadline, except where documented medical or compassionate grounds apply. The term project is completed in pairs and is assessed on the quality of the schema design and the clarity of the written justification, not on the volume of code produced.

Office hours. The instructor holds office hours on Tuesday and Thursday afternoons in Room 3521 and answers questions on the course forum within one working day. Teaching assistants run an additional consultation session before each assignment deadline. Students are encouraged to bring partially working queries rather than waiting until they have a complete solution, since most of the value of a consultation lies in the diagnosis rather than the answer.

Textbook. The recommended textbook is Database System Concepts by Silberschatz, Korth and Sudarshan. The PostgreSQL manual is treated as required reading for the chapters on indexing and transactions, and specific sections are listed on the weekly schedule. No purchase is necessary: a copy of the textbook is held on reserve in the university library and the manual is freely available online.

Academic honesty. All submitted work must be the student's own. Discussion of concepts between students is encouraged and expected, but the sharing of written answers, query text or project code is not permitted and is treated as plagiarism under the university's academic integrity regulations. Use of a generative language model to produce submitted answers must be disclosed in the submission. Suspected cases are referred to the departmental committee without exception, and the penalties range from a zero on the assessment to failure of the course.